

Lebanese University – Faculty of Engineering III
Sensors & Instrumentation Project
Individual Contribution Report

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Project: Smart Building Security and Fire-Protection System

Contribution Summary

Throughout the Smart Building Security and Fire-Protection System project, my primary responsibility was the complete development of the Arduino firmware. I wrote and implemented every code file in our repository, following the modular structure documented in the project README: `arduino_firmware.ino`, `entry_security.ino`, `fire_protection.ino`, `tcrt5000_sensor.ino`, `load_cell_sensor.ino`, `flow_sensor.ino`, `dht11_sensor.ino`, `config.h`, and `wifi_comm.ino`. I built each sensor driver step by step, ensuring that the optical (TCRT5000), force (load cell), flow, and temperature/humidity (DHT11) modules were each isolated into their own file and integrated cleanly through the main firmware and configuration header, in line with the modular firmware requirement.

A major part of my work was the calibration of every sensor. I developed the calibration routines for each modality, derived calibration curves from multiple reference points, and stored the resulting calibration constants in the dedicated configuration file rather than hard-coding them inline. As the project progressed, I revisited and refined these calibration routines further, improving measurement accuracy and reducing error margins across all five sensors after the initial testing phase.

I was also responsible for collecting the experimental data. I recorded the sensor readings from the Serial Monitor and organized them into the project's CSV dataset (`whole_dataset_for_the_sensors_project.xlsx`), ensuring all measurements were properly timestamped and formatted for analysis, consistent with the CSV logging requirement of the project.

On the hardware side, I took part in the physical wiring and installation of the system, connecting the sensors, actuators, and microcontroller into the final assembled circuit. I also prepared the circuit design in Fritzing, producing the schematic used for our hardware documentation.

In terms of documentation and communication, I worked on designing our presentation slides and contributed to finalizing the final technical report, helping express and organize our team's work clearly for submission. Finally, I participated in every workshop, meeting, and team session held throughout the three weeks of the project, working closely with my colleagues from the initial planning stage through to the live demonstration.

Areas of Contribution

- Complete Arduino firmware development (all `.ino/.h` files, modular sensor drivers)
- Sensor calibration: routines, calibration curves, and constants for all five modalities
- Ongoing calibration refinement to reduce measurement error
- Data acquisition and CSV dataset creation from Serial Monitor output
- Circuit wiring, hardware installation, and system assembly

- Circuit schematic design in Fritzing
- Presentation design for the 10-minute technical presentation
- Contribution to finalizing the final technical report
- Participation in all workshops, meetings, and team sessions

Declaration

I certify that the activities described above accurately reflect my individual contribution to the Smart Building Security and Fire-Protection System project.